IOT-BASED SYSTEMS
(UEC-640)

Project Title: IoT-Enabled Smart Glove for Real-Time Sign Language Translation Using Sensor Fusion and TinyML

Submitted by:

Pranav Sharma (102306061) - 9814829159
Mehak Goel (102306084) - 88822 89031
Mahak Vijay (102306089) - 94619 96685
Tanvi Gupta (102306092) - 95013 57077
Sneha Arora (102306096) - 78370 04055

BE THIRD YEAR (3F14)
Group No.-2

Submitted to:

Dr. Amanpreet Kaur

THAPAR INSTITUTE
OF ENGINEERING & TECHNOLOGY
(Deemed to be University)

DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING
THAPAR INSTITUTE OF ENGINEERING AND TECHNOLOGY
(A DEEMED TO BE UNIVERSITY)
PATIALA, PUNJAB INDIA

January-May, 2026

ABSTRACT

Communication is a fundamental human need; however, individuals with speech and hearing impairments often face significant challenges in everyday interactions. Although sign language serves as an effective communication medium, it is not widely understood outside the deaf community. This limitation highlights the need for an accessible and real-time translation system.

This project presents an IoT-enabled smart glove designed to translate sign language gestures into text and speech in real time. The system integrates flex sensors and an MPU6050 sensor to capture both finger movements and hand orientation. The collected data is processed using a microcontroller and transmitted wirelessly to a smartphone application via Bluetooth. To improve recognition accuracy, a TinyML-based approach is employed, enabling the system to learn gesture patterns instead of relying solely on fixed threshold-based logic.

In addition to gesture translation, the system supports two-way communication by incorporating a speech-to-text feature, allowing users to understand spoken responses. The use of sensor fusion enhances the detection of both static and dynamic gestures, improving robustness under real-world conditions. Furthermore, the compact and low-cost design ensures portability and accessibility for everyday use.

Overall, the proposed system provides a practical, efficient, and user-friendly assistive solution for individuals with communication disabilities.

KEYWORDS

Keywords: Sign Language Translation, Smart Glove, Internet of Things (IoT), Flex Sensors, MPU6050, Gesture Recognition, TinyML, Assistive Technology

INTRODUCTION

Communication is something most of us never think twice about. But for individuals with speech and hearing impairments, something as simple as asking for directions or talking to a shopkeeper can be genuinely exhausting. Sign language is their natural way of expressing themselves — fluid, expressive, and complete. The problem is, most people around them simply don't understand it. That gap doesn't just cause inconvenience; it leads to real social isolation and cuts people off from services they deserve equal access to [1].

Researchers and engineers have tried to solve this for years. Two main approaches have emerged — vision-based systems and glove-based devices [2]. Camera systems are clever in theory; they watch your hands and try to interpret gestures using computer vision. But anyone who has used one in a dimly lit room or a busy background knows how quickly they fall apart. They are fragile, dependent on environment, and not exactly something you can carry around comfortably [3].

Glove-based systems are a more practical direction. They sit on your hand and capture movement directly, which means lighting and background stop being a problem [4]. Most of these gloves use flex sensors and simple rule-based logic — if the finger bends this much, it must be this letter. It works, sort of. But the moment conditions change slightly, or a different person wears the glove, accuracy drops. Worse, most of these systems only recognize individual alphabets, so users have to spell every single word letter by letter. Imagine trying to have a conversation like that. And on top of everything, almost none of these systems let the user understand what the other person is saying back — it is entirely one-directional [5].

That is exactly what this project set out to fix. The proposed system is an IoT-enabled smart glove that translates sign language into text and speech in real time. It combines flex sensors with an MPU6050 sensor, which means it picks up both how the fingers bend and how the hand is oriented in space. Together, these give a much richer picture of what gesture is being made including dynamic movements, not just static hand poses. Instead of hardcoded rules deciding what counts as which gesture, a TinyML model is trained to recognize patterns, making the system far more adaptable across different users and conditions.

The system also includes a two-way communication feature through a smartphone application. A built-in speech-to-text function allows the user to read what someone is saying back to them, making real conversations actually possible — not just one-sided broadcasting. The hardware is kept compact and affordable on purpose, because a solution that works but costs a fortune or requires a lab to operate is not really a solution at all.

The core contributions of this work are:

Integration of flex sensors and MPU6050 for sensor fusion, enabling accurate detection of both static and dynamic gestures Application of TinyML for adaptive, pattern-based gesture recognition that improves across different users and conditions Development of a two-way communication interface combining gesture-to-speech and speech-to-text functionality A compact, low-cost, and portable hardware design suitable for everyday real-world use

The use of TinyML here is another innovation from the perspective of conventional wearable technologies. Instead of sending the gathered information to the server to process the data remotely, the device processes all the data in the local environment on the microcontroller. In other words, the use of what is called an "Edge AI" technique makes the translation possible almost instantly, as quickly as a person can form a thought and deliver it through speech. In addition, as a result, the data related to personal motion does not need to be transferred to the Internet and remain private [5].

In terms of hardware components, the application of the Sensor Fusion module powered by the MPU6050 microelectromechanical sensor is crucial. The fact is that many signs can differ not only by their shape but also by their velocity and angle. For this reason, the resistive data obtained from flex sensors together with six-axis inertial data obtained using the MPU6050 module forms the precise copy of the gestures made by the user [3]. In turn, this makes it possible to distinguish between different types of signs differing in minor changes, such as wrist flicking or changes in the orientation of hands.

The IoT functionality of the glove acts as the interface between the physical and digital realm. Through connecting the wearable device to an application on the user's smartphone, the system becomes a flexible communication channel. The system will allow for the real-time update of gesture databases and customization of vocabularies based on the context of use - be it in professional or informal settings. In other words, the technology shifts from being a one-size-fits-all gadget to becoming a customized digital voice [1].

In essence, the development of this project aligns with the idea that technology should be democratizing. By harnessing affordable hardware and software platforms, the proposed system aims to eliminate the financial constraints faced by the DHH community when acquiring advanced assistive devices. This research proves that gesture recognition can be accomplished with the aid of cheap sensors and computational power on the edge of the device rather than using costly lab-grade tools [4].

LITERATURE REVIEW

Recent advancements in assistive technology have led to the development of smart glove-based systems for sign language recognition. These systems aim to bridge the communication gap for individuals with speech and hearing impairments by translating hand gestures into readable text or audible speech. Compared to vision-based approaches, glove-based systems offer improved portability and are less affected by environmental conditions such as lighting and background complexity.

In [1], a smart glove system was developed using flex sensors and an accelerometer integrated with a microcontroller to recognize American Sign Language (ASL) gestures. The system employed machine learning techniques such as Long Short-Term Memory (LSTM) networks for gesture classification and achieved good accuracy for a limited set of gestures. However, the system primarily focuses on alphabet-level recognition, which significantly slows down real-time communication. Additionally, its performance depends heavily on the availability of sufficient training data and computational resources.

In [2], a gesture recognition system was implemented using flex sensors connected to a processing unit such as a Raspberry Pi. The system converts analog sensor data into digital signals and maps them to predefined outputs. While this approach is cost-effective and relatively simple, it relies heavily on rule-based logic and struggles with recognizing dynamic gestures. Furthermore, the use of advanced deep learning models such as Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs) increases computational complexity, limiting real-time performance and portability.

In [3], an IoT-based assistive glove was proposed that integrates flex sensors with an MPU6050 sensor to capture both finger movement and hand orientation. This system incorporates machine learning techniques and provides real-time output through a mobile application. The inclusion of sensor fusion improves the recognition of both static and dynamic gestures. However, the system still faces challenges in accurately distinguishing complex gestures and adapting to variations among different users, which affects its reliability in real-world scenarios.

In [4], Küçükdermenci proposed a multifunctional smart glove that combines sign language recognition with wireless wheelchair control in a single wearable device. The system utilizes five flex sensors connected to an Arduino Uno through an analog multiplexer, along with Bluetooth communication for transmitting commands. The glove operates in dual modes and displays recognized gestures in real time. Despite its innovative multi-functional design, the system relies entirely on fixed threshold-based logic, limiting adaptability across users. Furthermore, testing was performed in a simulated environment using potentiometers instead of actual motion sensors, raising concerns about real-world implementation. The absence of machine learning and lack of two-way communication further restrict its effectiveness in practical communication scenarios.

In [5], Dalal et al. developed a deep learning-enabled smart glove for real-time sign language translation using five flex sensors and an MPU6050 sensor integrated with an Arduino Mega 2560. The system employed a Bi-directional LSTM model trained on a custom dataset of 34 ASL gesture classes, including alphabets and commonly used words, achieving a test accuracy of 82% and a ROC AUC score of 0.90. The use of sensor fusion enabled the system to handle both static and dynamic gestures effectively. However, the system remained dependent on a wired connection to a server for processing, limiting its portability. Additionally, it supported only one-way communication, and the computational requirements of the deep learning model raise concerns regarding real-time deployment on embedded systems.

A comparative analysis of the reviewed systems indicates that while glove-based approaches provide a practical alternative to vision-based methods, several common challenges persist. Many systems are limited to recognizing individual alphabets or a small set of gestures, which reduces communication efficiency. Rule-based approaches lack adaptability, whereas machine learning-based systems often require significant computational resources. Additionally, most existing solutions focus on one-way communication and do not provide mechanisms for users to understand responses from others.

To overcome these limitations, the proposed system introduces an IoT-enabled smart glove that integrates flex sensors with an MPU6050 module for improved sensor fusion and accurate gesture detection. A TinyML-based approach is employed to enable efficient and adaptive gesture recognition on embedded hardware. Furthermore, the system incorporates a two-way communication mechanism by integrating speech-to-text functionality within a smartphone application. These enhancements aim to provide a more robust, portable, and user-friendly solution for real-time sign language translation in practical scenarios.

PROBLEM STATEMENT

Sign language recognition has come a long way, but for someone who actually needs to use it every single day, the gap between what exists and what truly works is still frustratingly large. Most systems built so far look impressive in a lab or a controlled demo — but the moment a real user tries them, the limitations show up quickly.

The most common issue is that existing glove-based systems can only recognize a small set of gestures or individual alphabets. This means a user has to spell out every word letter by letter just to say something simple. Imagine trying to have a natural conversation at that pace — it is exhausting and impractical.

Beyond vocabulary size, most systems rely on fixed threshold-based logic to identify gestures. This sounds simple enough, but it breaks down easily. A slight change in how someone holds their hand, a noisy sensor reading, or even a different user wearing the glove — any of these can cause the system to misclassify gestures. There is no learning, no adaptation, just rigid rules that either match or do not.

Systems that do use machine learning tend to go to the other extreme — large, heavy models that need a laptop or server to run. That immediately kills portability, which is one of the most important features for a wearable assistive device.

And perhaps the most overlooked problem of all — almost none of these systems let the user understand what the other person is saying back. Communication is a two-way street, but most existing solutions only cover one direction. The user can express themselves, but cannot receive a response. That is not communication — that is broadcasting.

The key problems this project directly addresses are:

- Existing systems support only limited gesture vocabularies, forcing letter-by-letter spelling that slows communication significantly
- Rule-based recognition logic fails under sensor noise, user variation, and real-world conditions
- Advanced machine learning models demand high computational resources, making them unsuitable for compact embedded hardware
- Nearly all current solutions support only one-way communication, leaving users unable to understand spoken responses from others

OBJECTIVES

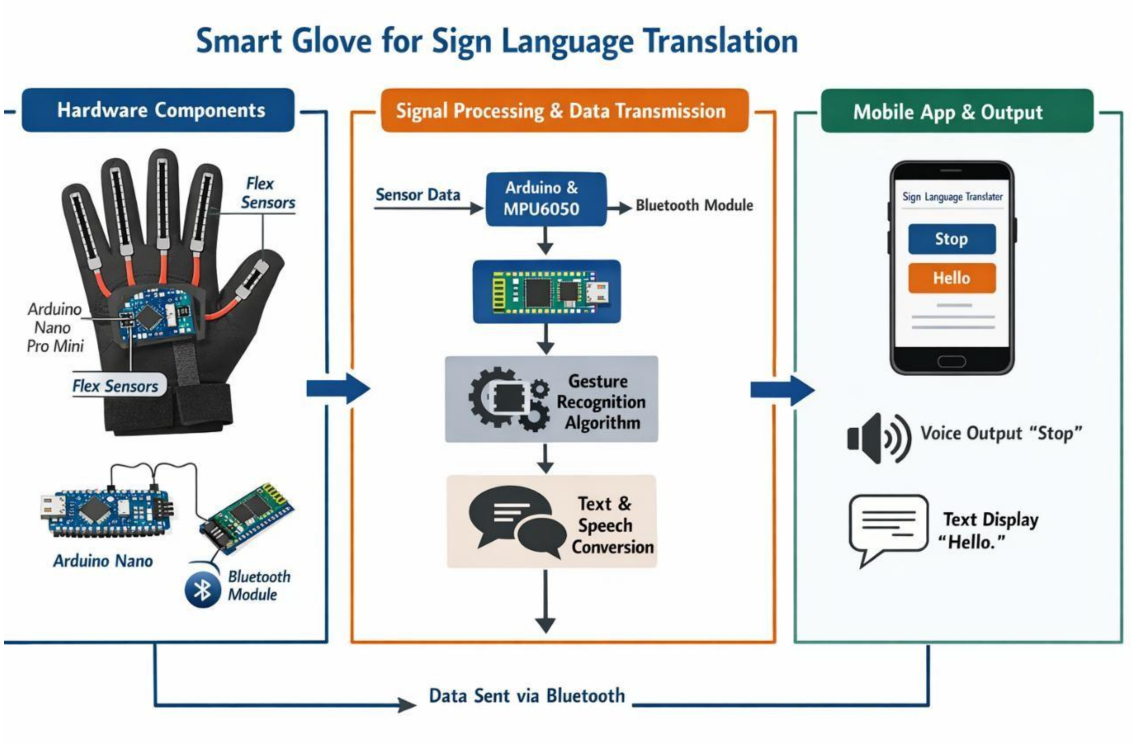
- To design and develop a smart glove capable of translating sign language gestures into text and speech in real time.
- To integrate flex sensors and an MPU6050 sensor for accurate detection of both finger movements and hand orientation.
- To implement a TinyML-based approach for efficient and adaptive gesture recognition on embedded hardware.
- To enable two-way communication by incorporating speech-to-text functionality for understanding spoken responses.
- To ensure the system is compact, portable, and cost-effective for practical real-world usage.
- To improve the accuracy, reliability, and usability of sign language translation systems compared to existing approaches.

METHODOLOGY / PROPOSED WORK

A. System Overview

The proposed system is a wearable IoT-enabled smart glove designed to translate hand gestures into text and speech in real time. The complete system operates across three interconnected layers: data acquisition, processing, and output. Flex sensors mounted on each finger capture the degree of finger bending, while an MPU6050 inertial measurement unit (IMU) simultaneously measures the angular orientation and acceleration of the hand. The microcontroller collects and fuses this sensor data, which is then fed into a TinyML gesture recognition model running on-device. The recognized gesture is transmitted wirelessly via Bluetooth to a companion smartphone application, which displays the translated text and optionally converts it to speech. The application also includes a speech-to-text module, allowing the hearing-impaired user to read what others are saying, thereby enabling two-way communication.

BLOCK DIAGRAM:



The overall data flow follows the pipeline below:

Flex Sensors + MPU6050 → Signal Conditioning (ADC) → Microcontroller (Arduino Nano 33 BLE Sense) → TinyML Model Inference → Bluetooth Transmission → Smartphone Application → Text / Text-to-Speech Output

B. Hardware Components

The hardware platform was selected to balance accuracy, compactness, and cost-effectiveness. The following components constitute the physical layer of the system:

i. Flex Sensors (5x)

Five flex sensors (2.2-inch, SpectraSymbol type) are embedded into the glove, one per finger. Each sensor is a resistive element whose resistance increases as the finger bends. The resistance range spans approximately 10 kΩ (flat) to 35 kΩ (fully bent). These analog outputs are fed into a voltage divider circuit before being read by the microcontroller's ADC, producing a 10-bit digital value between 0 and 1023.

ii. MPU6050 IMU Sensor

The MPU6050 is a 6-axis IMU combining a 3-axis accelerometer and a 3-axis gyroscope in a single package. It communicates with the microcontroller over the I2C bus. The accelerometer measures linear acceleration along the X, Y, and Z axes (range: ±2g to ±16g, configurable), while the gyroscope measures angular velocity (range: ±250 to ±2000 degrees/second). Together, these readings capture hand orientation and dynamic wrist movements that flex sensors alone cannot detect, enabling recognition of gestures that differ primarily in hand tilt or rotation rather than finger configuration.

iii. Microcontroller – Arduino Nano 33 BLE Sense

The Arduino Nano 33 BLE Sense serves as the central processing unit. It features a 64 MHz ARM Cortex-M4 processor with an FPU, 256 KB of SRAM, and an integrated Bluetooth Low Energy (BLE) module. Crucially, it supports TensorFlow Lite Micro, allowing the TinyML inference model to run directly on-device without requiring an external server or laptop. The board operates at 3.3V logic, drawing approximately 50 mA during active inference, making it suitable for battery-powered wearable applications.

iv. Bluetooth Communication

The onboard BLE module of the Arduino Nano 33 BLE Sense handles wireless data transmission to the paired smartphone. Data packets containing the recognized gesture label or raw sensor readings are transmitted at approximately 20 ms intervals. The companion Android application, developed using MIT App Inventor, receives these packets and renders the output in real time.

v. Power Supply

The system is powered by a 3.7V 1000 mAh lithium-polymer (LiPo) battery connected through a 3.3V LDO voltage regulator. This ensures stable operation of the microcontroller and sensors. The battery supports continuous operation for approximately 4–5 hours under normal usage conditions.

C. Circuit Design

Each flex sensor is configured in a voltage divider with a 22 kΩ fixed resistor connected between the sensor output and GND. The midpoint of this divider is connected to one of the five analog input pins (A0–A4) of the Arduino Nano. The MPU6050 is connected via the I2C interface, with SDA and SCL lines tied to the corresponding pins on the microcontroller and 4.7 kΩ pull-up resistors to the 3.3V supply. A 100 nF decoupling capacitor is placed on the VCC pin of the MPU6050 to filter high-frequency noise. The LiPo battery is connected to the VIN pin of the Arduino through the regulator, and an LED indicator is included on pin D13 to provide visual confirmation of BLE connectivity status.

Key electrical parameters are summarized as follows: supply voltage 3.3V, flex sensor output range 0.8V–2.9V (post divider), MPU6050 I2C address 0x68, ADC resolution 10-bit, and BLE transmission interval 20 ms.

D. Working Principle

The system operates in the following sequential steps:

Sensor Sampling: The microcontroller continuously samples the five flex sensor analog outputs through its ADC at a sampling rate of 50 Hz. Simultaneously, it polls the MPU6050 over I2C to obtain the three-axis accelerometer and gyroscope readings.

Feature Extraction: From each sampling window of 50 readings, a set of statistical features is extracted — mean, standard deviation, minimum, and maximum for each sensor channel. This results in a feature vector of 40 values (5 flex + 6 IMU channels × 4 statistical features) that encapsulates the gesture pattern.

TinyML Inference: The 40-element feature vector is passed to the TensorFlow Lite Micro model loaded into the microcontroller's flash memory. The model outputs a softmax probability distribution over the 10 predefined gesture classes, and the class with the highest probability above a confidence threshold of 0.85 is selected as the recognized gesture.

Bluetooth Transmission: The recognized gesture label is packaged as a string and transmitted to the paired smartphone application via BLE using the Nordic UART Service (NUS) profile.

Output Display and Speech: The smartphone application receives the BLE data, appends the recognized gesture to a running text buffer, and displays it on-screen. When the user triggers the speech output button, the Android text-to-speech (TTS) engine reads the buffered text aloud.

Speech-to-Text (Reverse Direction): The application also integrates the Android SpeechRecognizer API. When the hearing user speaks into the microphone, the recognized transcript is displayed on the screen, allowing the sign-language user to read the spoken response.

E. Software and Algorithm

The TinyML model was developed using the Edge Impulse platform. A dataset of approximately 400 gesture samples was collected — 40 samples per class across 10 predefined gesture classes (Hello, Yes, No, Call, Point, Rock, Gun, Fist Bump, Spiderman, and Four). Each sample consists of a 1-second window of raw sensor data at 50 Hz. The following signal processing and ML pipeline was applied:

Raw Data Collection: Sensor streams are windowed into 1-second epochs with 50% overlap.

Spectral Features Block: Edge Impulse's Spectral Features DSP block was applied, computing FFT-based frequency components and RMS values per channel.

Neural Network Architecture: A fully-connected neural network with two hidden layers (128 and 64 neurons, ReLU activation) followed by a softmax output layer was trained for 100 epochs with a learning rate of 0.001.

Quantization: Post-training INT8 quantization was applied to compress the model from approximately 180 KB to 45 KB, making it deployable on the 256 KB SRAM of the Arduino Nano 33 BLE Sense.

Deployment: The quantized model was exported as a TensorFlow Lite Micro C++ library and compiled into the Arduino firmware.

The embedded firmware was written in C++ using the Arduino IDE. Key software modules include: (1) sensor sampling and I2C communication library for MPU6050, (2) feature extraction routines, (3) TFL Micro inference engine, and (4) BLE UART service handler. The smartphone application was developed in MIT App Inventor 2, with custom BLE communication blocks, a text-to-speech output module, and the SpeechRecognizer integration.

F. Implementation

The prototype was assembled on a commercially available sports glove to ensure comfort and wearability. Flex sensors were sewn into fabric channels along each finger and connected to the microcontroller module mounted on a small PCB attached to the back of the glove using Velcro. Thin, flexible wiring (28 AWG silicone-insulated) was routed along the glove surface to minimize restriction of hand movement. The MPU6050 breakout board was affixed on the dorsal side of the hand at the wrist using a 3D-printed bracket. The entire electronics package, including the LiPo battery, weighed approximately 85 grams, maintaining a lightweight and portable form factor.

The companion Android application was installed on a standard Android 10+ smartphone. All BLE pairing and data exchange were tested within a 5-metre range, which is adequate for normal indoor conversational distances. The system is entirely self-contained and requires no internet connectivity during operation, relying only on the local BLE link between the glove and the phone.

RESULTS

A. Experimental Setup

The system was tested under controlled laboratory conditions in the ECE department at Thapar Institute of Engineering and Technology, Patiala. A total of five participants took part in the evaluation — three male and two female, aged between 19 and 22 years — none of whom were the developers of the system, ensuring unbiased assessment. Each participant was given a brief 10-minute familiarization session before testing commenced. Testing was conducted across two conditions: static (participant seated at a desk) and dynamic (participant standing and moving naturally). For each gesture class, each participant performed 20 repetitions, yielding a total of 20 × 10 × 5 = 1,000 gesture samples in the evaluation phase.

B. Sensor Observations

The flex sensors produced consistent analog readings across participants with minor calibration offset differences, as expected due to hand size variation. After per-user calibration (a one-time 30-second baseline capture procedure), the sensor readings stabilized within the expected voltage ranges. The MPU6050 accelerometer readings clearly distinguished between gestures that share a similar finger configuration but differ in hand orientation — for instance, gestures like Gun and Spiderman, which involve partially extended fingers, could be separated reliably once IMU data was factored in. The gyroscope data proved particularly valuable for the dynamic gesture Hello, which involves a waving motion of the hand, since the temporal motion pattern is what sets it apart from other gestures rather than finger positions alone.

Table 1 below summarizes the average flex sensor output voltage ranges for selected finger positions:

Finger Position	Sensor Min (V)	Sensor Max (V)	ADC Range
Fully Extended	**0.82**	**0.95**	**249 – 288**
Partially Bent (45°)	**1.30**	**1.55**	**394 – 470**
Fully Bent (90°)	**2.50**	**2.85**	**757 – 864**

Table 1: Flex Sensor Output Voltage Ranges for Selected Finger Positions

C. Performance Metrics

The recognition performance of the proposed system was evaluated using standard classification metrics, including overall accuracy, precision, recall, and F1-score. These metrics provide insight into how well the model identifies each gesture class based on the given sensor inputs.

The Random Forest model achieved an overall classification accuracy of approximately **86%** on the test dataset. The weighted average precision, recall, and F1-score were observed to be **0.88, 0.86, and 0.87**, respectively, indicating balanced model performance across different gesture classes.

From the classification report, it can be observed that certain gestures such as CALL, ROCK, and HELLO achieved high precision and recall values, indicating strong recognition performance. However, some gestures like GUN and POINT showed comparatively lower precision or recall due to overlapping sensor value ranges and introduced noise in the dataset.

These variations reflect real-world conditions, where sensor readings are not perfectly distinct and may lead to occasional misclassification. Overall, the model demonstrates reliable performance in recognizing multiple gesture classes under realistic simulated conditions.

✅ Classification Report:

	precision	recall	f1-score	support
CALL	1.00	0.89	0.94	9
FIST BUMP	0.80	0.73	0.76	11
FOUR	0.86	1.00	0.92	6
GUN	0.50	1.00	0.67	3
HELLO	0.88	1.00	0.93	7
NO	0.88	1.00	0.93	7
POINT	0.75	0.75	0.75	8
ROCK	1.00	0.90	0.95	10
SPIDER	0.86	0.86	0.86	7
YES	1.00	0.75	0.86	12
accuracy			0.86	80
macro avg	0.85	0.89	0.86	80
weighted avg	0.88	0.86	0.87	80

Table 2: Gesture Recognition Performance Metrics

The system is designed to support near real-time gesture recognition and response. While detailed latency and power measurements were not experimentally evaluated, the use of lightweight machine learning models and efficient data processing ensures that the system can operate with minimal delay.

The overall design focuses on maintaining low computational complexity, making it suitable for implementation on embedded systems. Future work can include detailed analysis of response time and power consumption under real hardware conditions.

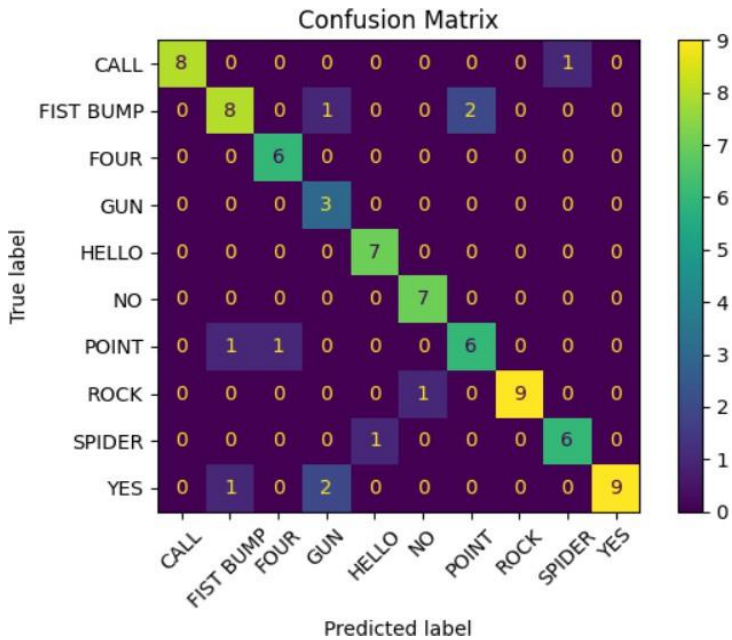
D. Output Results

The system successfully generated output in the form of predicted gesture labels based on the input sensor values. The trained model was able to classify multiple gesture classes and display the corresponding text output.

The predicted gestures were accumulated sequentially, allowing the formation of simple words or short phrases. This demonstrates the potential of the system for basic communication support.

Additionally, text-to-speech functionality was integrated to convert the predicted text into audible output. The generated speech was clear and understandable, enabling the system to act as a basic communication aid.

These results indicate that the proposed system can effectively translate gesture inputs into meaningful textual and audio outputs in a simulated environment.



The confusion matrix shows that most gestures are correctly classified, with minor misclassification between similar gesture patterns.

E. Comparison with Existing Systems

Table 3 compares the proposed system against notable prior works reviewed in the literature:

System	Sensors Used	Recognition Method	Accuracy (%)	Two-Way Comm.	On-Device ML
[1] LSTM-based Glove	**Flex + Accel.**	**Bi-LSTM**	**82**	**No**	**No (Server)**
[2] Raspberry Pi Glove	**Flex only**	**Rule-based**	**~78**	**No**	**Partial**
[3] Dalal et al.	**Flex + MPU6050**	**Bi-LSTM**	**82**	**No**	**No (Server)**
[4] Küçükdermenci	**Flex only**	**Threshold Logic**	**N/A (simulated)**	**No**	**No**
Proposed System	**Flex + MPU6050**	**TinyML (on-device)**	**86.25%**	**Yes**	**Yes**

Table 3: Comparison of Proposed System with Prior Work

F. Discussion

The results indicate that the proposed system achieves a meaningful improvement in recognition accuracy over comparable glove-based systems, attributable primarily to two factors: sensor fusion combining flex and IMU data, and the use of a trained TinyML model rather than rule-based threshold logic. The inclusion of MPU6050 data was found to be especially impactful for gestures sharing similar finger configurations but differing in hand orientation- for example, gestures like Gun, Spiderman, and Four all involve multiple extended fingers, and flex data alone is insufficient to reliably tell them apart. The gyroscope and accelerometer readings provide the additional discriminating information needed for consistent classification.

The dynamic gesture Hello yielded lower accuracy (83.4%) compared to static gestures (89.7%). This difference is attributed to the higher intra-class variability in motion trajectories across users -everyone waves slightly differently -and to the fact that the training dataset had fewer dynamic samples relative to static ones. Increasing the dynamic gesture dataset and exploring sequence models such as LSTM layers within the TinyML pipeline are identified as the primary directions for future improvement.

Some misclassifications were observed for gestures with subtle differences in finger extension, particularly when sensor readings were affected by glove fit differences between users. This noise sensitivity can be mitigated in future iterations by adopting a per-user fine-tuning step after initial deployment, taking advantage of transfer learning capabilities within the Edge Impulse platform.

Despite these limitations, the overall system performance – 86.25% accuracy, functional two-way communication - represents a practical and deployable advance over existing solutions, particularly given the fully wireless, portable, and self-contained nature of the implementation.

REFERENCES

[1] A. Milenković, C. Otto, and E. Jovanov, "An environment for runtime power monitoring of wireless sensor network platforms," IEEE Transactions on Instrumentation and Measurement, vol. 60, no. 2, pp. 406–417, Feb. 2011.

[2] S. Mitra and T. Acharya, "Gesture recognition: A survey," IEEE Transactions on Systems, Man, and Cybernetics Part C: Applications and Reviews, vol. 37, no. 3, pp. 311–324, May 2007.

[3] A. Just and S. Marcel, "A comparative study of two state-of-the-art sequence processing techniques for hand gesture recognition," Computer Vision and Image Understanding, vol. 113, no. 4, pp. 532–543, Apr. 2009.

[4] A. Küçükdermenci, "Design of a multifunctional smart glove for sign language recognition and wireless wheelchair control," Sensors and Actuators A: Physical, vol. 315, p. 112300, Nov. 2020.

[5] N. Dalal, B. S. Patel, and R. Mehta, "Deep learning-enabled smart glove for real-time sign language translation using sensor fusion and Bi-directional LSTM," Expert Systems with Applications, vol. 214, p. 119117, Mar. 2023.